# Sahil Pathak

AI/ML and full-stack engineer

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## SUMMARY

Final-year computer science undergraduate specialising in AI and data science, with deep learning research and production full-stack work. Co-developed a PyTorch multimodal model that reached 91.24% accuracy in depression detection, and cut a production MERN app's load time by 50 to 80%. Core stack: Python, JavaScript, TypeScript, React, Next.js, Node.js, TensorFlow, Hugging Face.

## EXPERIENCE

**Research intern, Vellore Institute of Technology (VIT)** May 2026 to Jul 2026

- Analysed 20+ research papers (2020 to 2026) and led end-to-end assembly of a model fusing text, EEG, wearable and audio/video data.
- Co-developed the PPEMDD framework in PyTorch, pairing a gated fusion network (DGN) with multi-head cross-attention (MHCA) to perform three tasks in one model: depression classification, PHQ-9 severity regression and DSM-5 symptom recognition.
- Achieved 91.24% accuracy, 79.23% recall and a 0.7809 F1-score on a 2,250-sample held-out test set (Reddit and WU3D corpora), using a leakage-aware protocol with no test-set threshold tuning.
- Engineered availability-aware fusion that handles missing modalities.

**Software developer, AMG Turnkey Projects LLP** Jan 2025 to Jun 2025

- Owned development of the live production website (React, Vite, Tailwind CSS, Node.js, Express, MongoDB), built to make the company's construction services findable across three states.
- Worked with the operations team for six months to deliver civil accounting modules and REST APIs for project workflows.
- Diagnosed performance bottlenecks and optimised the application, cutting load time by 50 to 80% and improving reliability.

## RESEARCH

**PPEMDD**, an IEEE-format research paper (2026) proposing an adaptive multimodal deep learning framework for depression detection with gated cross-modal attention fusion.

## PROJECTS

**Sarkari Saathi, voice AI for government schemes** TypeScript, React, Express, Gnani Prisma and Timbre, LLM

- Built a voice-first app, for Gnani AI's Great Indian AI Internship Challenge, that finds the government schemes a person may qualify for from a conversation in Marathi, Hindi or English. Live at sarkari-saathi-kjgb.onrender.com.
- The LLM only extracts facts as zod-validated JSON; TypeScript rules checked against official pages decide eligibility across 13 schemes, so it cannot invent one. 4 to 6 questions, about 3 s per spoken turn, 76 tests.

**AI code editor (NeuraCraft)** Next.js, FastAPI, scikit-learn, Convex, Clerk

- Built a full-stack AI code editor (Monaco) with real-time inline suggestions for four Python ML libraries. Live at ai-compiler-eta.vercel.app.
- Trained a scikit-learn model to predict the ML framework in use, served through FastAPI on Render with a Next.js client on Vercel.
- Implemented Clerk authentication and Convex real-time storage; curated a growing coding-error dataset to retrain the model.

**Desi Totes e-commerce platform (client project)** React, Node.js, Express, MongoDB, Razorpay

- Launched a live store for a client with Razorpay payments in two currencies (INR and USD) and persistent carts. Live at desitotes.com.

**AI image generator** Python, PyTorch, Hugging Face Diffusers, Transformers

- Fine-tuned and deployed a diffusion image-generation pipeline, speeding up inference by 25% and reducing memory use by 30%.

## EDUCATION

**B.Tech, Computer Science and Engineering, DY Patil International University, Pune** Aug 2023 to Aug 2027

**B.Sc. (Hons.), Data Science and Artificial Intelligence, IIT Guwahati** Aug 2023 to Aug 2027

Coursework: data structures and algorithms, OOP, DBMS, operating systems, computer networks, machine learning.

## SKILLS

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| **Languages** | Python, JavaScript, TypeScript, SQL, C++, C, HTML, CSS |
| **AI and ML** | Machine learning, deep learning, NLP, multimodal learning, generative AI, diffusion models, model evaluation |
| **ML libraries** | PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, Diffusers, Pandas, NumPy |
| **Web** | React, Next.js, Node.js, Express, FastAPI, Flask, Redux, Tailwind CSS, Vite, REST APIs |
| **Data and tools** | MongoDB, Firebase Firestore, Convex, Git, GitHub, Postman, VS Code, Vercel, Render, CI/CD |

## ACHIEVEMENTS

- Smart India Hackathon: top 15 teams in the college round, in a six-member team.
- FOSS Hackathon 2025: collaborated on an open-source AI coding assistant that improved automated code correction.
- SharkIndia (Avenyou Club): won a prize for an AI-based solution.
- Deloitte Australia technology job simulation (Forage, May 2025): assessed a client scenario and proposed a custom dashboard.
- Generative AI certification.